Spectral Earth Mini: leakage-free crop classification and label-free mineral mapping

GeoMine intern final round · technical report · all numbers produced by *python scripts/reproduce_all.py*; headline numbers are spatial-split, mean ± sd over 3 seeds.

1. Data and preprocessing

Scenes. Indian Pines corrected (145×145×200, 16 classes, AVIRIS June 1992, ~20 m), Salinas corrected (512×217×204, 16 classes, AVIRIS 1998, 3.7 m) and Cuprite f970619t01p02_r02_sc03 reflectance (512×614×224, AVIRIS 1997) from the EHU scene page; the live server refused scripted downloads (HTTP 403), so the script fetches the Internet Archive's byte-exact capture of the same URLs and checks SHA-256 plus the published class counts. **Wavelengths.** The .mat files carry none. Indian Pines: centres and FWHM from 92AV3C.spc, the Purdue/MultiSpec calibration file of the original 220-band image, with EHU's removed bands (104–108, 150–163, 220) dropped. Salinas: splib07's AVIRIS-Classic 1998 table with EHU's removed bands (108–112, 154–167, 224). Cuprite: the 1997 table exists only inside the 5.5 GB splib07 release (bot-protected), so the 1998 table is a *documented proxy*; every channel the 1997 product zero-fills lies at 1.36–1.39 / 1.83–1.93 µm in it (asserted in tests), and yearly drift (~1–2 nm) is small against the ~10 nm FWHM. The AVIRIS centre sequence steps backwards at the spectrometer joins, so no code assumes a sorted axis. **Bands and normalisation.** The corrected crop cubes already exclude the water-vapour windows; we keep the remaining bands (permutation importance in §6 shows the 1.33 and 1.97–1.99 µm shoulders carry information) and expose *drop_ranges_nm* for further removal. Per-band z-scoring is fitted on training pixels only (values span orders of magnitude across bands, which would otherwise dominate the RBF distance); the CNN additionally uses 30 PCA components fitted on training pixels only.

2. Split design

Random: stratified 10 % of each class to train, rest test; tuning uses a stratified 20 % hold-out of the training pixels. **Spatial:** the scene is tiled into 15×15 px blocks, each wholly assigned to train/val/test. Out of 1,500 seeded random assignments we keep the one closest to 60/20/20 by labelled pixels, subject to every class with ≥50 px being in test (only label *counts* are used, never model output). A 3-px buffer equal to the CNN patch radius (7×7 patches, receptive field = patch) removes training pixels within 3 px of any test or validation block, and validation pixels within 3 px of a test block, so **no training input patch contains a test pixel**. *verify_spatial()* re-checks block purity, the patch condition and class coverage from the final masks at run time and in the test suite (which also injects a leak and expects it to be caught). Realised shares (Indian Pines seed 0): train 0.57 / val 0.20 / test 0.23 (4643/1610/1832 px; 2164 labelled px dropped by the buffer). Each seed redraws the split, so seed variance includes split variance. **Why 15 px:** Indian Pines fields span few blocks, so coverage of test forces some classes out of training at larger sizes: 0/0/0 classes (≥50 px) missing from train at 15 px, 0/1/2 at 20 px and 3/4/3 at 30 px (seeds 0/1/2). 15 px (300 m) is the smallest allowed size, so it is the most permissive for leakage between blocks; the buffer and the block-size study (§5) address that.

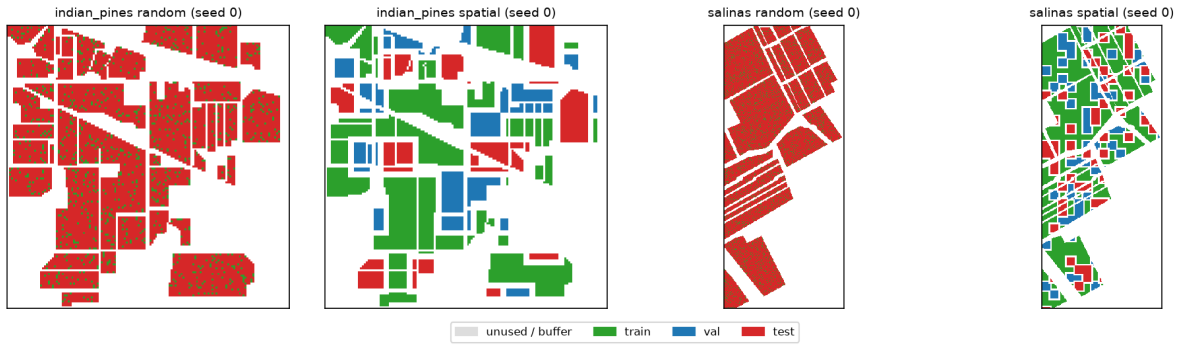


Fig. 1. Seed-0 split maps. Spatial splits show whole blocks and the white buffer rings around test/val blocks.

3. Models

SVM: RBF-SVM on all z-scored bands; C ∈ {1,10,100,1000} × γ ∈ {0.001,0.005,0.02} chosen on validation OA; Platt-scaled probabilities; Salinas training capped at 8,000 stratified pixels. **CNN3D**: three 3-D convolutions (8/16/32 filters, kernels 7×3×3, 5×3×3, 3×3×3, BatchNorm, ReLU) on 1×30×7×7 PCA patches, then FC-128 → dropout 0.4 → softmax; 96,176 parameters. Adam, weight decay 1e-4, flips/rotations, learning rate ∈ {1e-3, 3e-3} and the checkpoint chosen on validation OA (validation loss alone stopped at epoch 2 because of over-confidence); temperature scaling fitted on validation logits. Training takes 55 s (Indian Pines) and 350 s (Salinas) per seed on an 8-thread CPU; SVM 21 s. Both families use identical splits and the same tuning rule (validation only).

4. Results

scene	split	model	OA	AA	κ	macro-F1	ECE raw	ECE temp.	train px	test px
indian pines	spatial	svm	0.636 ± 0.042	0.682 ± 0.037	0.578	0.620	0.120	–	4674	1638
indian pines	spatial	cnn3d	0.616 ± 0.035	0.619 ± 0.028	0.552	0.562	0.285	0.076	4674	1638
salinas	spatial	svm	0.929 ± 0.015	0.949 ± 0.005	0.921	0.944	0.015	–	8000	8593
salinas	spatial	cnn3d	0.969 ± 0.024	0.977 ± 0.014	0.965	0.975	0.021	0.012	25638	8593
indian pines	random	svm	0.801 ± 0.005	0.656 ± 0.005	0.771	0.670	0.109	–	1025	9224
indian pines	random	cnn3d	0.934 ± 0.007	0.908 ± 0.015	0.924	0.927	0.012	0.009	1025	9224

salinas	random	svm	0.934 ± 0.002	0.965 ± 0.002	0.927	0.967	0.023	–	5415	48714
salinas	random	cnn3d	0.991 ± 0.001	0.995 ± 0.001	0.990	0.995	0.002	0.001	5415	48714

Table 1. Test metrics, mean ± sd over seeds 0–2 (spatial rows are the headline). ECE: 15 equal-width bins on max probability; 'temp.' = CNN after temperature scaling on validation.

On the honest split the classical SVM (0.636 ± 0.042 OA) is level with the CNN (0.616 ± 0.035); the CNN's spatial context, which wins by 13 points on the random split, does not transfer to unseen fields. Raw CNN probabilities are badly over-confident on the spatial split (ECE 0.285); temperature scaling on validation reduces this to 0.076 (the map/app use the scaled values). Weakest Indian Pines classes (SVM, spatial, F1): Alfalfa 0.00, Grass-pasture-mowed 0.00, Corn 0.37, Soybean-notill 0.44. The corn/soybean tillage variants are spectrally near-identical early-season canopies on different soils; their confusions dominate the matrices (appendix).

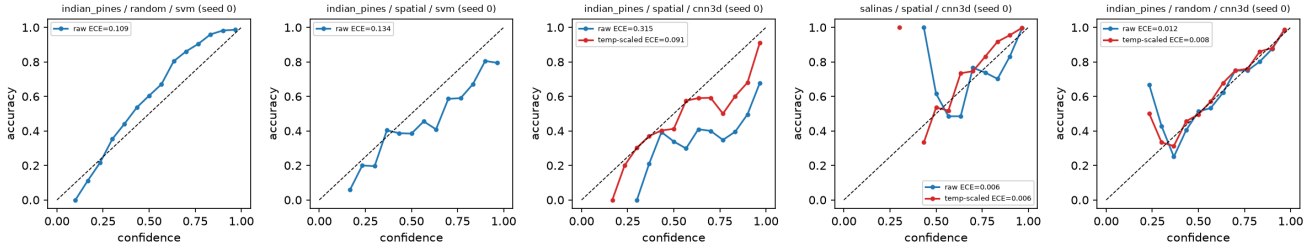


Fig. 2. Reliability diagrams (seed 0). Points below the diagonal = over-confidence.

5. The random-vs-spatial gap

Size. On Indian Pines, OA drops from 0.801 to 0.636 for the SVM (–16.5 points, κ 0.77→0.58) and from 0.934 to 0.616 for the CNN (–31.8 points), even though the spatial split has 4.6× more training pixels. On Salinas the CNN drops only 2.2 points. **Mechanism 1: test pixels are neighbours of training pixels.** Under the random split 54 % of test pixels touch a training pixel (median distance 1 px); under the spatial split none do (median 8 px). **Mechanism 2: neighbours are near-duplicates.** The mean spectral angle between same-class pixels grows from 1.9° at 1 px to 3.7° at 32 px and 4.3° for same-class pixels >40 px apart (different classes: 9.1°). A random-split model is therefore largely matching a test pixel to the field it came from (field-specific soil, tillage, illumination), not to its crop. The CNN is hit hardest because its 7×7 patch overlaps training patches directly. **Direct evidence:** within the random split the CNN's accuracy falls from 0.97 for test pixels adjacent to a training pixel to 0.80 at 4–6 px; for the spatial models it falls from 0.73 (4–6 px) to 0.49 (11–20 px, SVM). Accuracy is a function of distance to training data, and the random split simply never measures the far end. **Block size.** SVM OA is 0.636 ± 0.041 at 15 px, 0.641 ± 0.045 at 20 px and 0.462 ± 0.058 at 30 px (doubling the block); the CNN at 30 px reaches 0.411. At 30 px whole fields, and with them 3–4 classes, disappear from training, so part of that drop is missing classes, not only distance. **Salinas** stays high; our hypothesis is that its fields are large relative to a 15-px (55 m) block, so test blocks often sit inside fields that also feed training, which would mean block size should be set in metres relative to field size rather than in pixels. Doubling to 30 px (111 m) lowers Salinas SVM OA only to 0.890 and CNN to 0.959, consistent with but not proof of this; a field-level split is the real test (§9).

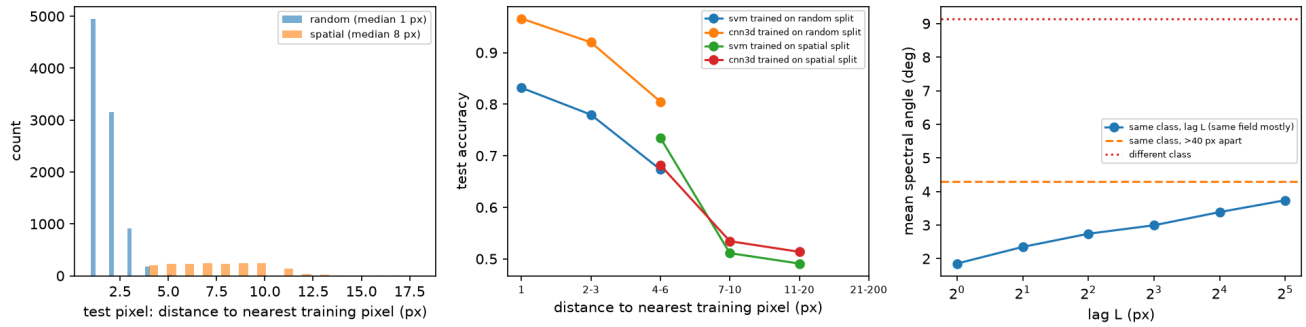


Fig. 3. Left: distance from test pixels to the nearest training pixel. Middle: test accuracy vs that distance (seed-0 models). Right: spectral autocorrelation.

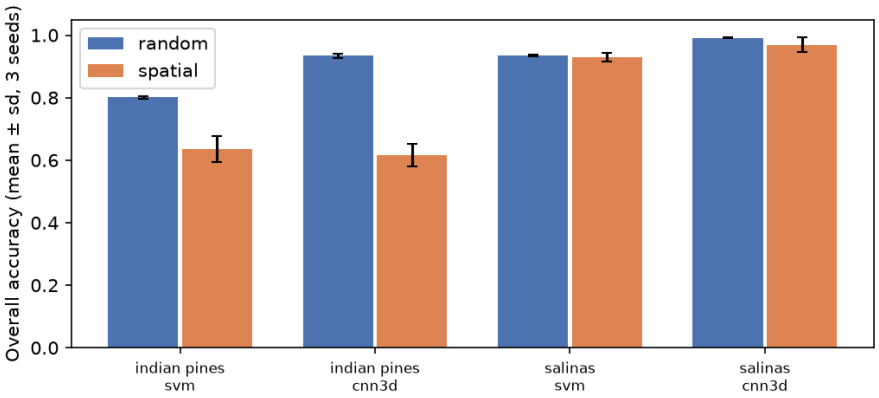


Fig. 4. OA by split and model (error bars = sd over 3 seeds).

6. Which wavelengths matter

Per-band permutation importance of the spatial-split SVM, measured on each seed's validation pixels (never test), ranks these centres highest (nm, 3-seed mean): 725, 716, 1983, 696, 1973, 706, 627, 735, 1333, 2341, 697, 637. Physical reading: (i) **chlorophyll red edge** 696–735 nm dominates both per-band and 5-band-window importance (the window version peaks at 764–812 nm, the NIR shoulder set by canopy cover and leaf structure); with corn/soy at <5 % cover in June, red-edge position and NIR level separate canopy density and residue cover. (ii) **Visible 577–637 nm**: chlorophyll/carotenoid absorption and soil colour. (iii) **Leaf/atmospheric water**: 1158 nm (1.2 µm leaf-water band) and the shoulders of the 1.38 and 1.9 µm water-vapour windows (1332, 1973–1993 nm), which respond to canopy water and surface moisture; the 970 nm feature did not rank high here. (iv) **Cellulose/lignin/soil** near 2.0–2.35 µm (2043, 2341 nm): dry residue and soil minerals, which differ between tillage classes. Per-band importances are small and noisy because adjacent bands substitute for each other. ≤**15 bands.** Top-15 by importance (≥30 nm apart) gives 0.529 ± 0.105 OA, greedy forward selection with an LDA scorer on validation 0.473 ± 0.084, 15 random bands 0.563 ± 0.041, versus 0.636 ± 0.041 with all 200. Selection on a small, spatially-correlated validation set does not beat random bands: the picks over-fit the validation fields (seed-to-seed sd ≈ 0.1). **Sentinel-2-like** bands (average of AVIRIS bands inside each S2 band; 12 bands: B1, B2, B3, B4, B5, B6, B7, B8, B8A, B9, B11, B12; B10 unavailable because its 1.36–1.39 µm range was removed with the water bands) reach 0.656 ± 0.037, i.e. as good as the full spectrum under the honest split. Broad averaged bands remove noise and fine field-specific detail that does not generalise between fields; the extra hyperspectral detail helps the random split far more than the spatial one.

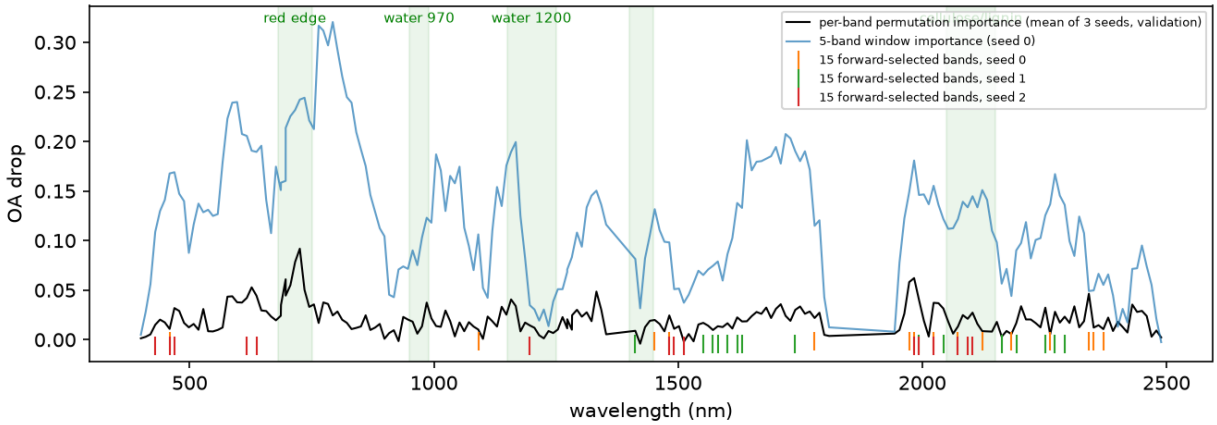


Fig. 5. Permutation importance (validation) vs wavelength; ticks = forward-selected bands per seed. Shaded: physics regions.

Transfer to Salinas (same code, same config)

The pipeline ran unchanged (204 bands handled by the band table and PCA). Spatial CNN 0.969 ± 0.024, SVM 0.929 ± 0.015. What broke or changed: (1) the 15-px block is only 55 m at 3.7 m GSD, far smaller than Salinas fields, so the spatial split is much weaker evidence here than on Indian Pines (see §5); (2) the RBF-SVM does not scale to 25k training pixels, so it had to be capped at 8,000; (3) one CNN seed had a markedly worse split (sd 0.024); (4) the hardest classes are Vinyard_untrained (0.82), Grapes_untrained (0.91), Corn_senesced_green_weeds (0.97): Grapes_untrained vs Vinyard_untrained are both trellised vines on bare soil, separable mainly by context; (5) the crop/non-crop mask for index maps had to be redefined because the class sets differ.

Index maps. Each index uses the AVIRIS band whose centre is nearest the nominal wavelength; R<nm> is that band's value. Indian Pines: NDVI = (R802.5 - R666.6) / (R802.5 + R666.6); NDRE = (R792.9 - R715.8) / (R792.9 + R715.8); REP = 696.5 + (744.7 - 696.5) x (Rre - R696.5) / (R744.7 - R696.5), Rre = (R666.6 + R783.3) / 2 [linear 4-point method]. Salinas: NDVI = (R798.1 - R673.4) / (R798.1 + R673.4); NDRE = (R788.5 - R721.3) / (R788.5 + R721.3); REP = 702.1 + (740.5 - 702.1) x (Rre - R702.1) / (R740.5 - R702.1), Rre = (R673.4 + R778.9) / 2 [linear 4-point method]. The crop-class maps mask pixels not predicted as crop classes by the CNN3D class map. Inputs are at-sensor values, not surface reflectance, so index values are relative between fields, not absolute.

7. Mineral mapping and its validation (Cuprite)

Library. Native-resolution splib07a spectra (alunite ×2, kaolinite ×2, muscovite, buddingtonite, calcite, chalcedony; non-targets montmorillonite, hematite, jarosite, dickite, opal) were convolved by us with Gaussian SRFs (FWHM = channel spacing) at the adopted centres, because no 1997 AVIRIS convolution could be retrieved. Against USGS's own AVIRIS-1998 convolution the RMS difference over 2.0–2.45 µm is ≤0.0056 reflectance. **Physics baseline.** Upper-convex-hull continuum removal over 2.00–2.45 µm (45 bands), band depth and position at each diagnostic feature (alunite/kaolinite 2.165, muscovite 2.205, buddingtonite 2.12, calcite 2.34, chalcedony ~2.25 µm), then SAM on the absorption vector 1 – CR with a 0.02 minimum depth. The threshold (0.35 rad) is the smallest angle reaching 95 % of the best macro-F1 on labelled *synthetic* spectra, never tuned on the image. **Learned model.** An MLP (128-64) trained only on 21,000 synthetic spectra: target fraction 0.3–1 mixed with 1–2 backgrounds (hematite or featureless albedo 0.05–0.6 with slope), linear or intimate (Hapke two-stream single-scattering-albedo) mixing, illumination ×0.6–1.4 ± 0.01 offset, and per-band Gaussian noise from the image's own shift-difference noise estimate (median SNR 74 in the SWIR window). An explicit unknown class (backgrounds, non-target minerals, sub-detection fractions) plus rejection at p<0.5 gives the 'unknown/none' output. Held-out synthetic OA 0.984 (SAM 0.882); at 2× and 3× the estimated noise it falls to 0.942 and 0.868. These are *synthetic* numbers only.

Checks without labels. (1) *Agreement:* where SAM maps a mineral, the MLP gives the same label for 92 % of pixels; overall agreement 0.853, κ 0.38. κ is low because the MLP maps far more pixels (calcite 31,877 vs 8,217; muscovite 12,537 vs 3,806; kaolinite 5,203 vs 487): it was trained to call a mineral at 30 % abundance, while SAM demands a whole-shape match. (2) *Spectra and residuals:* SAM-mapped pixels fit the library feature with median R^2 0.80 (alunite), 0.84 (muscovite), 0.88 (calcite); median feature positions 2169, 2209 and 2338 nm match the expected 2165, 2205 and ~2340 nm (Fig. 6). (3) *Sensitivity:* ±33 % on the SAM threshold changes the mapped fraction from 0.9 % to 14.3 % of valid pixels; the depth gate (0.01–0.04) has no effect at this threshold. (4) *USGS map:* the published AVIRIS map (no colour key on the page, different extent and orientation) can only be compared qualitatively. Published Cuprite studies (Swayze, Clark et al.) describe two alteration centres with alunite and kaolinite cores, muscovite at their margins and carbonate in the hills to the west; our maps show a coherent alunite zone, muscovite along drainages and western carbonate, which is consistent with that description but not verified pixel by pixel (Fig. 7).

Where we believe the map is wrong. (a) *Kaolinite is under-mapped by SAM* (alunite and kaolinite share the 2.165 µm band, library angle 0.42 rad); alunite–kaolinite mixtures, which dominate the alteration centres, are forced into one label, so the SAM alunite zone probably contains kaolinite. (b) *Buddingtonite*: SAM maps none; the MLP's 191 pixels show a 2.02–2.06 µm dip but not the 2.12 µm NH_4 minimum, i.e. residual atmospheric CO_2 at the window edge, so they are false positives. (c) The MLP's large calcite areas are partly real carbonate but likely include other 2.33–2.35 µm absorbers absent from the library (dolomite, chlorite, epidote, Mg-OH clays). (d) Dickite was a non-target in training, so dickite-rich pixels are pushed to 'unknown' or kaolinite. (e) The 1998 wavelength proxy can shift feature positions by about one channel. (f) MLP probabilities are not calibrated on real data and must not be read as accuracy.

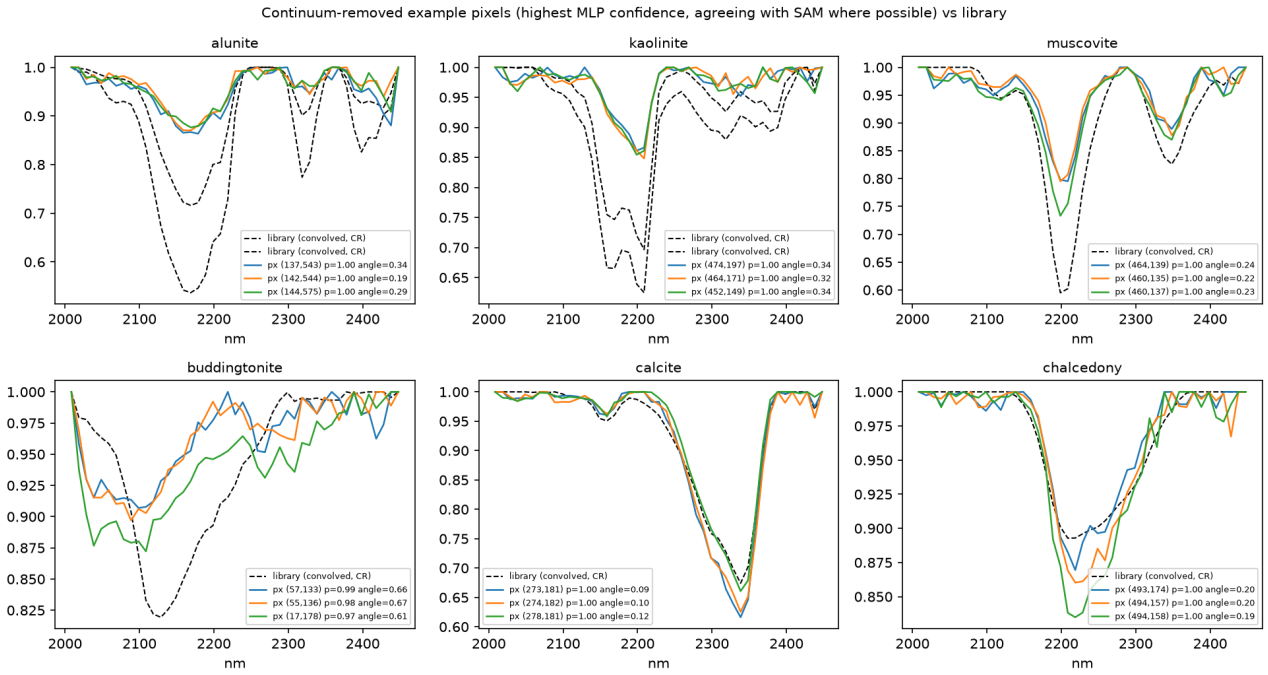


Fig. 6. Continuum-removed example pixels (highest MLP confidence, agreeing with SAM where possible) vs library spectra.

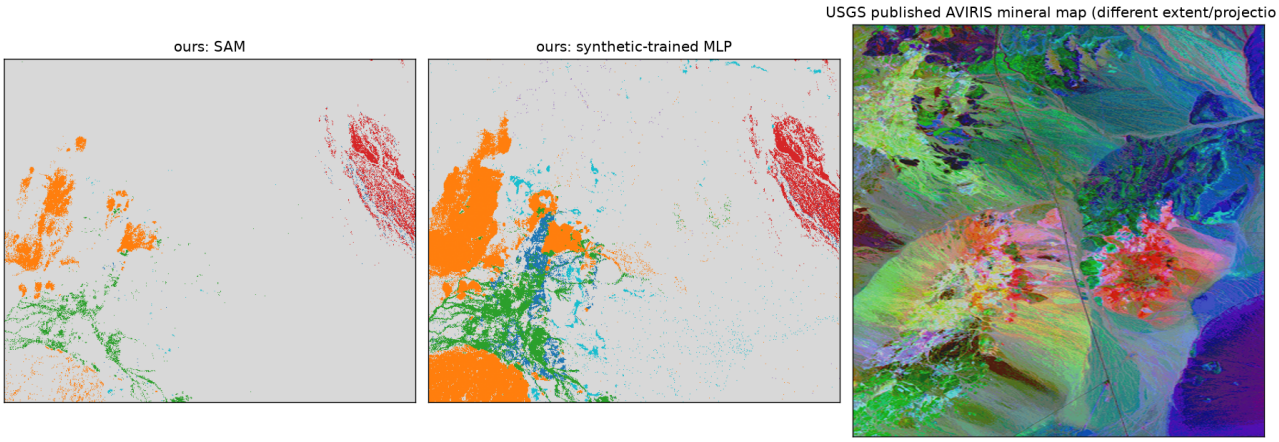


Fig. 7. SAM and MLP mineral maps beside the published USGS AVIRIS Cuprite map (qualitative only).

8. Limitations

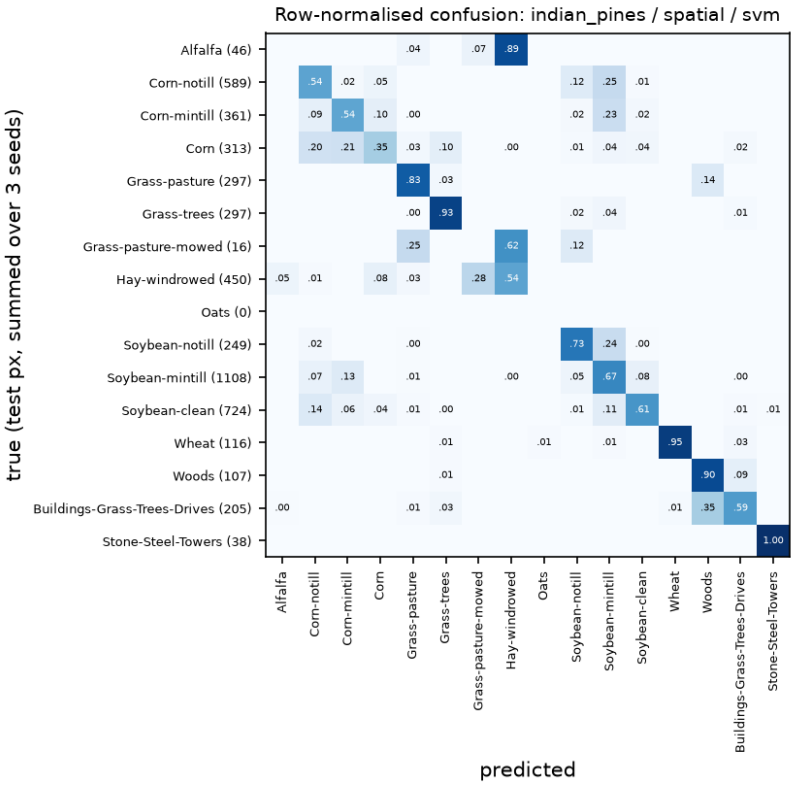
Indian Pines is a 145×145 scene: three seeds of block assignment explain much of the variance, and some classes sit in one or two fields, so per-class numbers rest on very few independent fields. The spatial split guarantees pixel-level separation, not field-level: a field cut by a block boundary can still feed both train and test beyond the 3-px buffer. The SVM is spectral-only, and its Salinas training set is capped. Hyper-parameter grids are small. The Cuprite wavelengths are a proxy, the library has a single sample for most minerals (grain size, crystallinity and solid-solution variation are not covered), and SAM threshold choice is only as good as the synthetic data. There is no field truth for Cuprite, so mineral 'accuracy' remains unknown. Maps are not georeferenced because the .mat files carry no geo-metadata. **Analysis choices made after seeing results** (disclosed for transparency): (i) the CNN checkpoint rule was changed from validation loss to validation OA after a seed-0 trial run whose test OA we had seen (the validation curve, which stopped at epoch 2, motivated it, but the test number was visible); (ii) the SAM threshold rule was changed to the 95 % knee after inspecting real-scene class counts (no labels involved); (iii) forward selection was added after the importance-based 15-band result was seen; both are reported. The default band set was *not* changed after an ablation dropping the near-water bands (IP SVM OA 0.648 vs 0.636, within seed sd).

9. With two more weeks

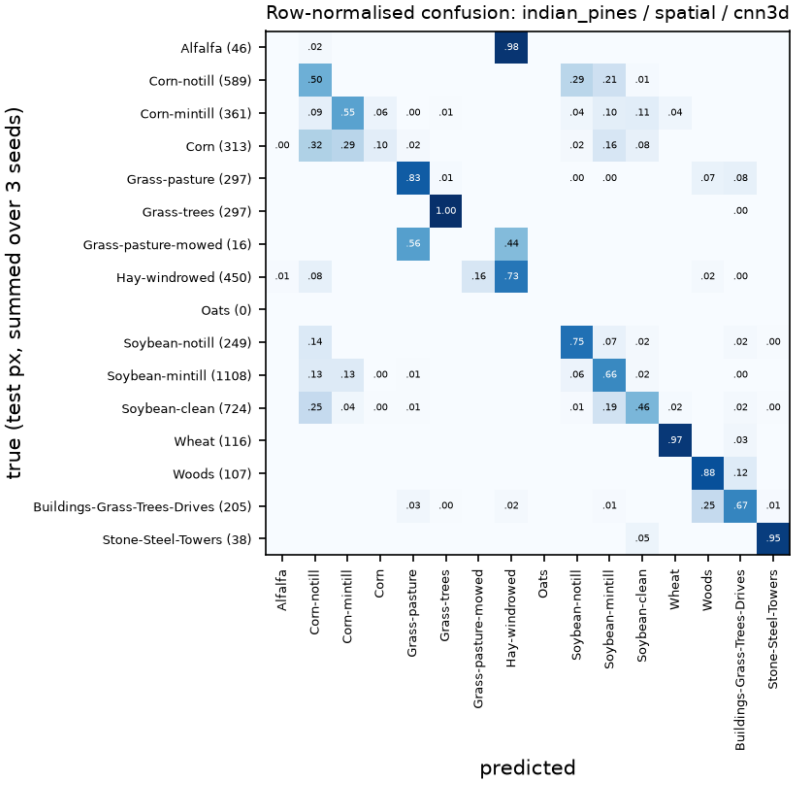
(1) Field-level splits: segment fields from the GT and assign whole fields, and a metre-based block for Salinas; report OA against block size and distance to training data as a curve. (2) A spatial-aware CNN regulariser and patch-level domain augmentation (brightness/soil mixing), plus spectral-only 1D-CNN to separate spectral from contextual gains. (3) Stability selection for bands (repeat selection over bootstrap of validation fields) and S2/EnMAP-band transfer. (4) Minerals: add dickite, montmorillonite, jarosite, dolomite as targets, multiple library samples per mineral, a Tetracorder-style multi-feature fit, mixture-aware (unmixing) abundances, and retrieve the 1997 AVIRIS channel table; georegister the USGS map and compute class agreement quantitatively. (5) Calibrate the mineral model's rejection with a small set of field-verified Cuprite sites from the literature.

Appendix

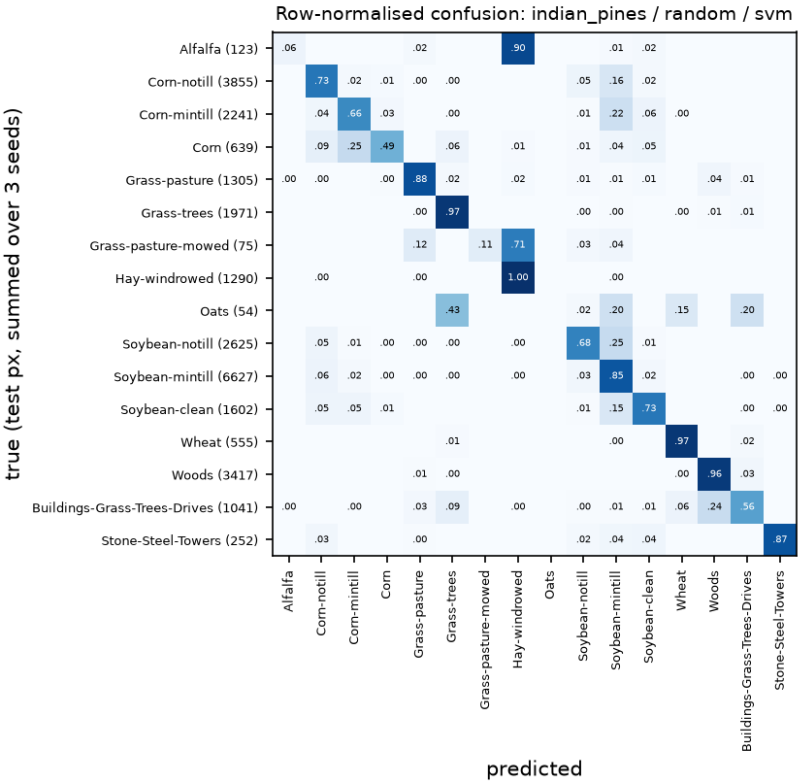
A1. Confusion matrices (row-normalised, test pixels summed over 3 seeds).



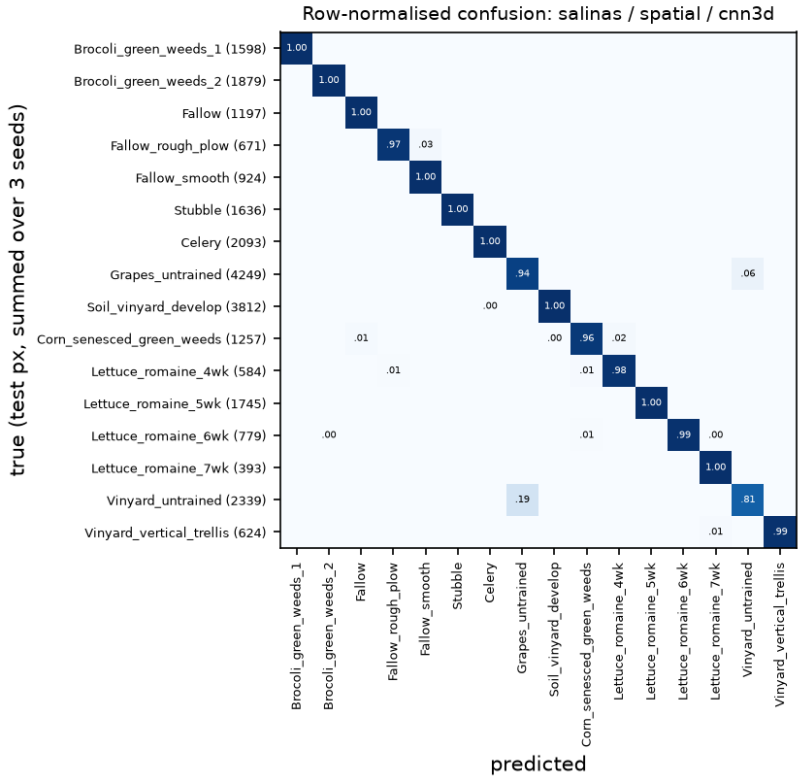
Indian Pines spatial SVM.



Indian Pines spatial CNN3D.



Indian Pines random SVM.



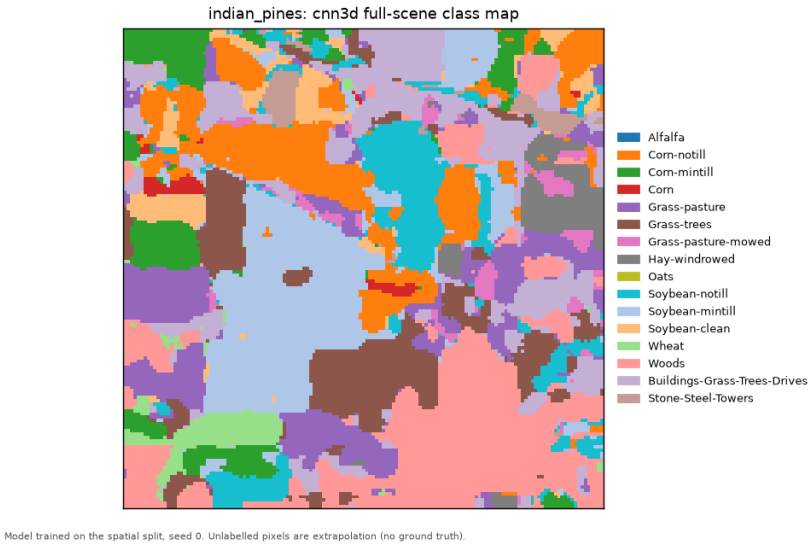
Salinas spatial CNN3D.

A2. Per-class F1 (mean over 3 seeds).

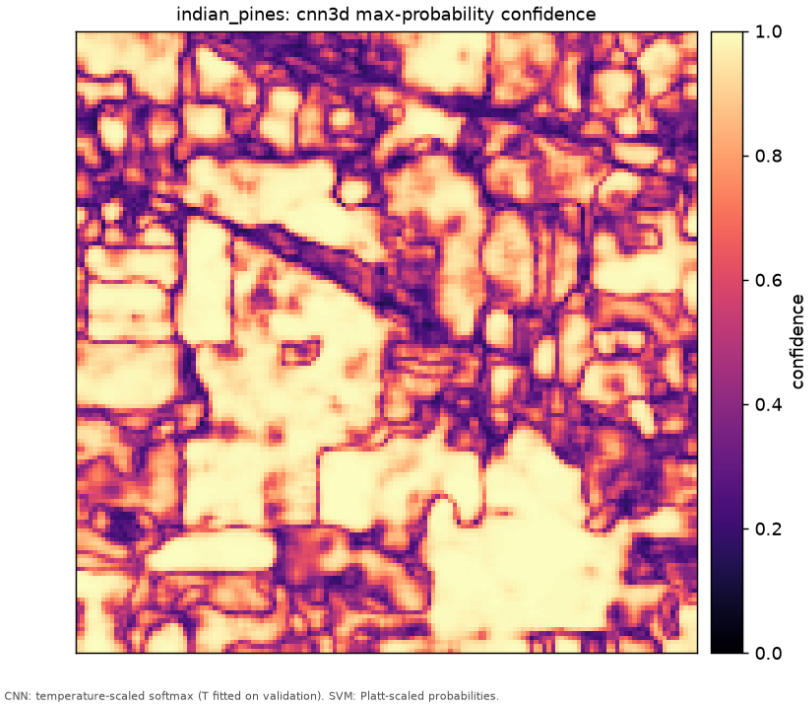
class	IP rand SVM	IP spat SVM	IP spat CNN	seeds in test (spat)	test px (spat s0)
Alfalfa	0.0999	0.0	0.0	1	46
Corn-notill	0.759	0.4614	0.3505	3	130
Corn-mintill	0.6989	0.4483	0.4435	3	153
Corn	0.5632	0.3676	0.1785	3	121
Grass-pasture	0.9047	0.7976	0.603	3	56
Grass-trees	0.9314	0.8839	0.9837	3	72
Grass-pasture-mowed	0.1784	0.0	0.0	1	0

Hay-windrowed	0.9267	0.6919	0.6855	3	356
Oats	0.0	n/a	n/a	0	0
Soybean-notill	0.7287	0.4397	0.3934	3	153
Soybean-mintill	0.7868	0.5733	0.5814	3	342
Soybean-clean	0.7235	0.6792	0.5613	3	201
Wheat	0.9237	0.9483	0.7721	3	60
Woods	0.9348	0.479	0.533	3	88
Buildings-Grass-Trees-Drives	0.6429	0.7338	0.6996	3	44
Stone-Steel-Towers	0.9119	0.9507	0.8877	3	10

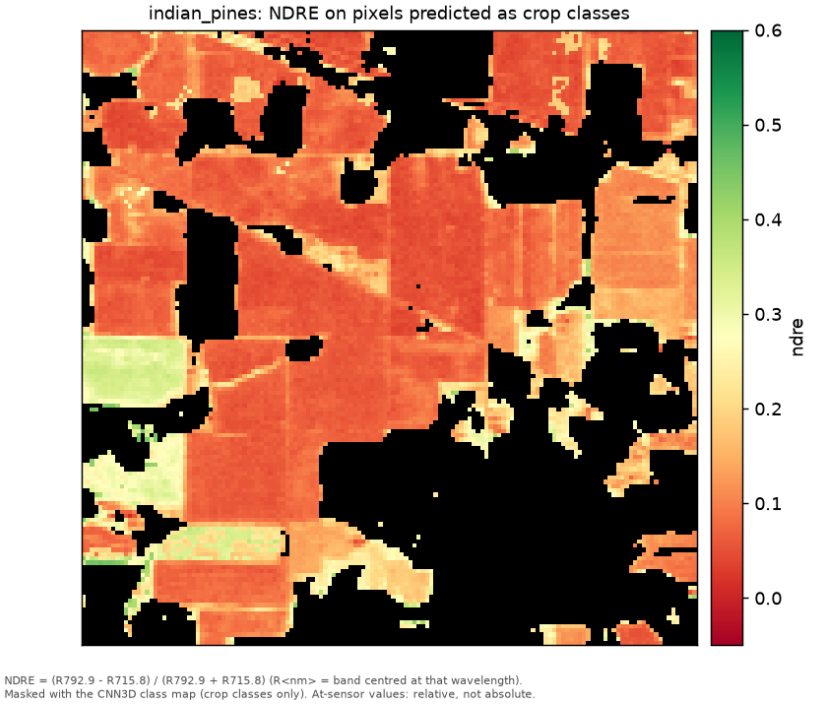
A3. Maps (full set in outputs/).



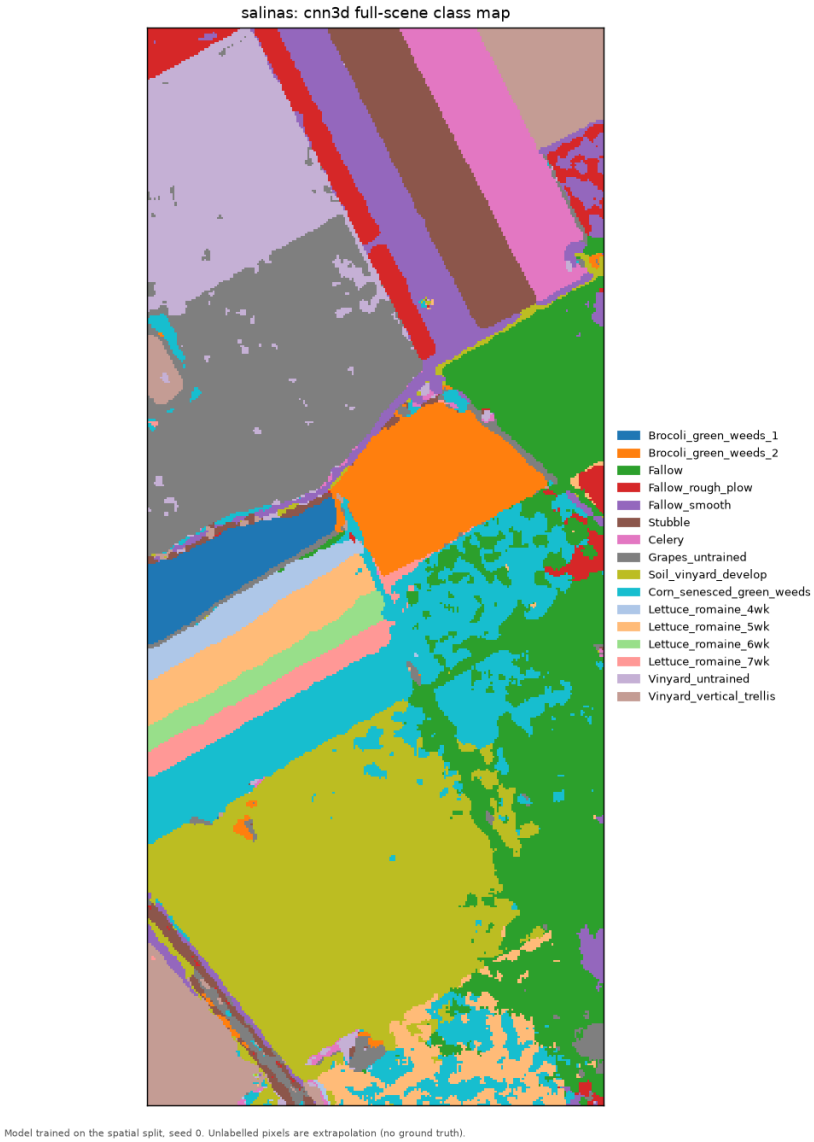
Indian Pines CNN3D class map.



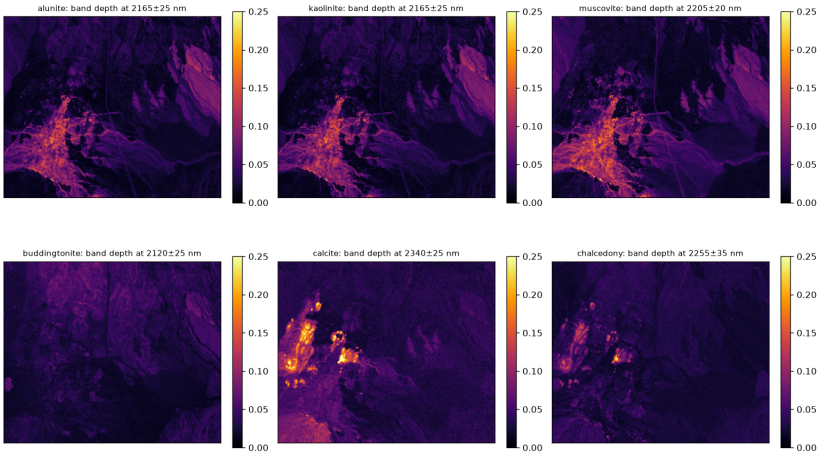
Indian Pines CNN3D confidence (temperature-scaled).



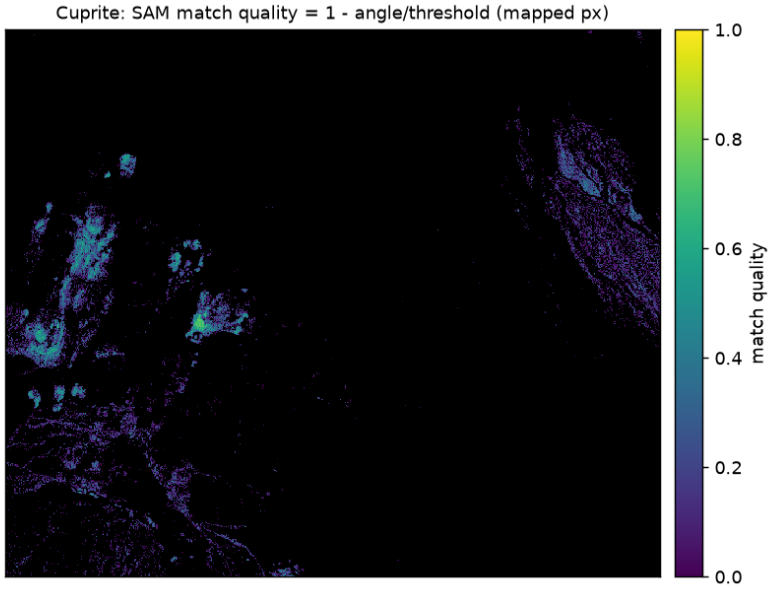
Indian Pines NDRE on predicted crop classes: NDRE = (R792.9 - R715.8) / (R792.9 + R715.8).



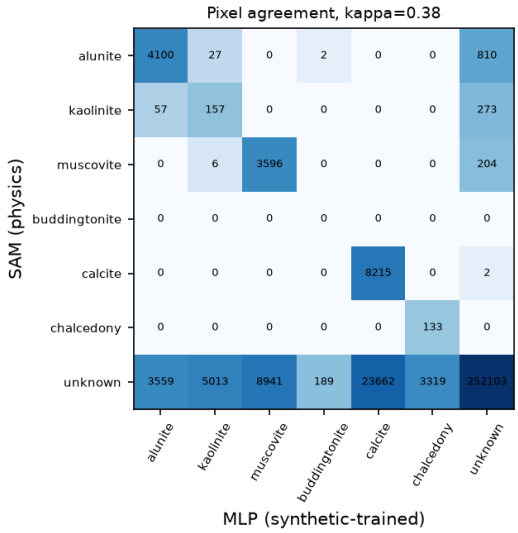
Salinas CNN3D class map.



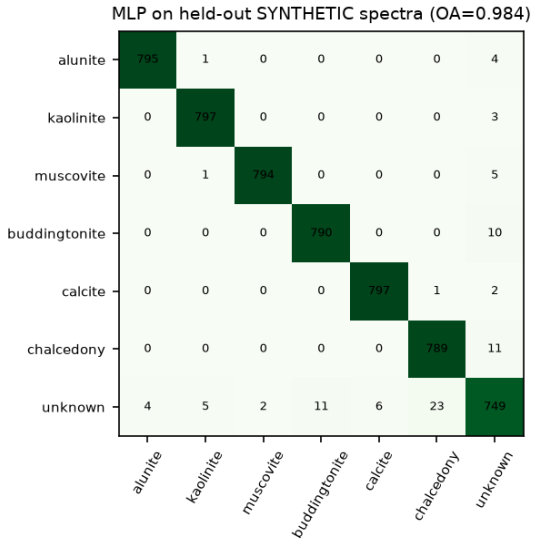
Cuprite diagnostic band-depth maps.



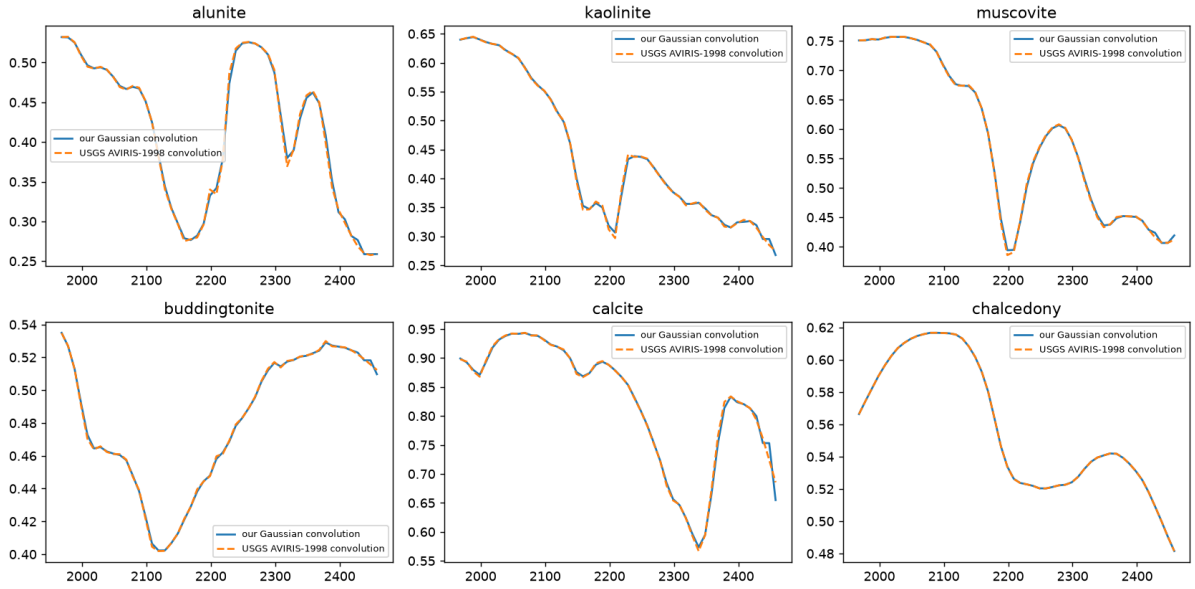
Cuprite SAM match quality.



SAM vs MLP pixel agreement.



MLP on held-out synthetic spectra.



Our Gaussian convolution vs USGS AVIRIS-1998 convolution.

A4. Reproducibility: seeds 0/1/2; configs/default.json; environment and per-run hyper-parameters in experiments/runs/*.json and outputs/reports/results_per_seed.csv; logs in experiments/logs/. AI assistance: see README.